

Spectral Solver Design — Liouville / Sturm–Liouville / Chebyshev Methods for the Stratified Pressure Equation

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0. About this document

This document consolidates four prior technical notes (`anelastic_SL_spectral_design.md`, `liouville_SL_spectral_derivation.md`, `reduced_pressure_liouville.md`, `liouville_singularity_causality.md`) produced during 2026-05-01..03 into a single integrated reference, updated with the post-Phase-0-ext+ conclusions. The four original files are retained in the repository as a record of the reasoning trajectory; each one now carries a prominent header pointing the reader here and to the formal report.

Relationship to the formal report. The formal English report `docs/spectral_stratified_poisson_report_2026-05-03.md` is the citation-ready document (18 pages, authored by Kiriko / Tsinghua University) that presents the Phase 0 ext+ conclusions and quantitative evidence. The present document is the **internal engineering design record**: it retains the full mathematical derivations, the discussion of implementation trade-offs, and the historical context, and is the intended starting point for a developer joining the project.

Three sibling documents:

- `docs/spectral_stratified_poisson_report_2026-05-03.md` — formal English report
- `docs/spectral_experiments.md` — experimental record (Phase 0, reduced-pressure, g-mode, polytropic-index convergence)
- `docs/singular_basis_survey_2026-05-02.md` — GYRE / Dedalus survey

Part I Motivation and Problem Setup

1. Motivation

The existing `pseudo_spectral` solver handles uniform-density, doubly-periodic, two-dimensional incompressible Navier–Stokes ($\rho = \text{const}$, $\nabla \cdot \mathbf{\Gamma} = 0$) in vorticity–streamfunction form. Stellar convection requires stratified density $\rho_0(y)$, which promotes the pressure equation to a **variable-coefficient elliptic problem**. Fourier modes are no longer eigenfunctions, and the standard $\mathcal{O}(N \log N)$ pseudo-spectral Poisson solve fails.

This document integrates two technical strategies:

1. **Liouville–SL basis (Parts II–III).** The substitution $\hat{p} = \sqrt{\rho_0} q$ reduces the variable-coefficient Poisson operator to a Schrödinger form whose Sturm–Liouville eigenfunctions are common to all k_x . The Poisson solve becomes a pipeline “cuFFT + GEMM + pointwise division + GEMM + cuFFT”.
2. **Direct Chebyshev collocation (Part IV).** The Phase 0 ext+ investigation demonstrates that, for the Eddington standard model ($n = 3, \sigma = 3$), raw Chebyshev collocation attains spectral convergence to machine precision without any Liouville-type substitution. **This is the Phase 1 main route.**

Both routes share the same Chebyshev-grid infrastructure; they differ only in how the operator is assembled. The SL route is retained as a Phase 2+ optional backend.

2. Problem formulation

2.1 Variable-density incompressible pressure equation

Dividing the variable-density incompressible momentum equation by ρ_0 and taking the divergence gives

$$\nabla \cdot \left(\frac{1}{\rho_0(y)} \nabla p \right) = f(\Gamma) \quad (2.1)$$

where $\rho_0(y)$ is the background stratified density, depending only on y . For the anelastic system $\nabla \cdot (\rho_0 \Gamma) = 0$, the same operator appears after the analogous projection; the mathematical structure is identical.

2.2 Fourier reduction in the homogeneous direction

The x direction remains homogeneous and periodic. Expanding $p(x, y) = \sum_{k_x} \hat{p}(k_x, y) e^{ik_x x}$ reduces (2.1) for each horizontal mode to a one-dimensional variable-coefficient ODE:

$$\frac{d}{dy} \left[\frac{1}{\rho_0(y)} \frac{d\hat{p}}{dy} \right] - \frac{k_x^2}{\rho_0(y)} \hat{p} = \hat{f}(k_x, y). \quad (2.2)$$

Issue. The $1/\rho_0(y)$ factor in front of the k_x^2 term couples SL modes. A direct eigenfunction expansion of $\frac{d}{dy}[(1/\rho_0)\frac{d}{dy}]$ alone is not enough: the k_x^2 term introduces dense mode coupling.

The two resolutions of this issue — the Liouville substitution (Parts II–III) and direct Chebyshev collocation (Part IV) — are what this document develops.

Part II Liouville Normal-Form Reduction

3. The Liouville substitution

3.1 Change of dependent variable

Set

$$\boxed{\hat{p}(y) = \sqrt{\rho_0(y)} q(y),} \quad (3.1)$$

and compute the transformed operator.

Proposition 1 (Liouville reduction). *Under the substitution (3.1),*

$$\frac{d}{dy} \left[\frac{1}{\rho_0} \frac{d\hat{p}}{dy} \right] = \frac{1}{\sqrt{\rho_0}} [q'' + W(y) q],$$

with the **Liouville potential**

$$W(y) = \frac{\rho_0''}{2\rho_0} - \frac{3(\rho_0')^2}{4\rho_0^2}.$$

Proof. Differentiating $\hat{p} = \sqrt{\rho_0} q$:

$$\begin{aligned} \hat{p}' &= \frac{\rho_0'}{2\sqrt{\rho_0}} q + \sqrt{\rho_0} q', \\ \frac{1}{\rho_0} \hat{p}' &= \frac{\rho_0'}{2\rho_0^{3/2}} q + \frac{q'}{\sqrt{\rho_0}}. \end{aligned}$$

Differentiating the second line:

$$\begin{aligned} \frac{d}{dy} \left[\frac{1}{\rho_0} \hat{p}' \right] &= I_1 + I_2, \\ I_1 &= \left[\frac{\rho_0''}{2\rho_0^{3/2}} - \frac{3(\rho_0')^2}{4\rho_0^{5/2}} \right] q + \frac{\rho_0'}{2\rho_0^{3/2}} q', \\ I_2 &= -\frac{\rho_0'}{2\rho_0^{3/2}} q' + \frac{q''}{\sqrt{\rho_0}}. \end{aligned}$$

The q' contributions cancel, leaving

$$I_1 + I_2 = \frac{1}{\sqrt{\rho_0}} [q'' + W(y) q]. \quad \square$$

□

3.2 The reduced equation

Substituting (3.1) and (3.2) into (2.2) and multiplying by $\sqrt{\rho_0}$:

$$\boxed{q'' + W(y) q - k_x^2 q = g(y), \quad g \equiv \sqrt{\rho_0} \hat{f}.} \quad (3.4)$$

Defining the **Liouville–Schrödinger operator**

$$\mathcal{T} \equiv \frac{d^2}{dy^2} + W(y), \quad (3.5)$$

(3.4) reads

$$[\mathcal{T} - k_x^2] q = g. \quad (3.6)$$

The key structural property: $\mathcal{T}$ does not depend on k_x . The horizontal wavenumber enters only as an additive shift of eigenvalues. A single precomputed spectral basis therefore serves all k_x simultaneously.

4. Spectral diagonalisation

4.1 The eigenvalue problem

Consider the Sturm–Liouville (equivalently, time-independent Schrödinger) eigenvalue problem

$$\mathcal{T} \psi_n = \psi_n'' + W(y) \psi_n = -\mu_n \psi_n, \quad n = 0, 1, 2, \dots \quad (4.1)$$

with appropriate boundary conditions (periodic, Dirichlet, or Neumann). Standard SL theory guarantees:

1. Eigenvalues $\{\mu_n\}$ are real and non-decreasing: $\mu_0 \leq \mu_1 \leq \mu_2 \leq \dots$.
2. Eigenfunctions $\{\psi_n\}$ form a complete orthonormal basis of $L^2([0, L_y])$.
3. When $W \equiv 0$ (uniform density), ψ_n reduces to the standard Fourier modes and $\mu_n = n^2 \pi^2 / L_y^2$.

Theorem 1 (Universal diagonalisation). *The eigenfunctions $\{\psi_n\}$ simultaneously diagonalise the operator $\mathcal{T} - k_x^2$ for every k_x :*

$$[\mathcal{T} - k_x^2] \psi_n = -(\mu_n + k_x^2) \psi_n.$$

4.2 Solution by eigenfunction expansion

Expanding $q = \sum_n a_n \psi_n$ and $g = \sum_n g_n \psi_n$ and substituting into (3.6), orthonormality gives

$$a_n(k_x) = -\frac{g_n(k_x)}{\mu_n + k_x^2}. \quad (4.3)$$

No mode coupling. The variable coefficient $1/\rho_0$ is entirely encoded in the eigendata $\{(\mu_n, \psi_n)\}$. Each (k_x, n) solve is a scalar division, structurally identical to the uniform-density Fourier–Poisson solve $\hat{p}(\Gamma) = -\hat{f}(\Gamma)/|\Gamma|^2$.

5. Complete SL-GEMM algorithm

5.1 Precomputation (once, or whenever ρ_0 changes)

1. Compute $W(y)$ from $\rho_0(y)$ via (3.3).
2. Solve the 1D Schrödinger eigenvalue problem (4.1) for $\{\mu_n, \psi_n\}$.
3. Form the transform matrix $\Psi_{in} = \psi_n(y_i)$, $\Psi \in \mathbb{R}^{N_y \times N_y}$.
4. Precompute the vector $\sqrt{\rho_0(y_i)}$.

5.2 Per-timestep Poisson solve

Step	Operation	Formula	Cost
1	FFT in x	$f(x, y) \rightarrow \hat{f}(k_x, y)$	$\mathcal{O}(N_x N_y \log N_x)$
2	Weight	$g = \sqrt{\rho_0} \cdot \hat{f}$	$\mathcal{O}(N_x N_y)$
3	Forward SL transform	$G = \Psi^\top g$	$\mathcal{O}(N_y^2 N_x)$
4	Pointwise divide	$Q_n(k_x) = -G_n / (\mu_n + k_x^2)$	$\mathcal{O}(N_x N_y)$
5	Inverse SL transform	$q = \Psi Q$	$\mathcal{O}(N_y^2 N_x)$
6	Weight	$\hat{p} = \sqrt{\rho_0} \cdot q$	$\mathcal{O}(N_x N_y)$
7	IFFT in x	$\hat{p}(k_x, y) \rightarrow p(x, y)$	$\mathcal{O}(N_x N_y \log N_x)$

Dominant cost: the pair of dense matrix–matrix products in steps 3 and 5, giving overall complexity $\mathcal{O}(N_y^2 N_x)$ per Poisson solve.

5.3 Complexity comparison

Method	y direction	Total ($N \times N$)	GPU characteristics
Fourier ($\rho_0 = \text{const}$)	division	$N^2 \log N$	cuFFT (optimal)
Chebyshev + tridiagonal	Thomas alg.	N^2	serial per column
SL-GEMM	GEMM	$\mathbf{\Gamma}^3$	cuBLAS batched (tensor cores)
Iterative (CG / MG)	SpMV $\times k$	$N^2 k$	convergence-dependent

While SL-GEMM has higher asymptotic complexity, dense matrix multiplication achieves near-peak GPU arithmetic throughput (high arithmetic intensity, fully parallel, tensor-core compatible). For $N_y \lesssim 4096$, batched DGEMM wall-clock time is competitive with latency-bound Thomas, and often faster.

Part III Reduced-Pressure Formulation

6. Motivation — a change of dependent variable weakens the singularity

The $1/\rho_0$ coefficient in (2.1) is introduced by the specific algebraic manoeuvre of dividing by ρ_0 before taking the divergence. **Changing the dependent variable changes the singularity.**

Define the **reduced pressure** $\pi \equiv p/\rho_0$ (specific enthalpy perturbation). Substituted into the anelastic momentum equation with the continuity constraint, this yields

$$\nabla \cdot (\rho_0 \nabla \pi) = \tilde{f}, \quad (6.1)$$

where $\tilde{f}$ absorbs the buoyancy, advection, and $\nabla \rho_0$ terms. **The structural change:** the elliptic operator in (6.1) has coefficient ρ_0 (not $1/\rho_0$). The density now appears in the **numerator**.

7. Reduced-pressure Liouville analysis

7.1 Fourier reduction

$$\frac{d}{dy} \left[\rho_0 \frac{d\hat{\pi}}{dy} \right] - k_x^2 \rho_0 \hat{\pi} = \tilde{f}(k_x, y). \quad (7.1)$$

7.2 Substitution $\hat{\pi} = \rho_0^{-1/2} q$

Direct symbolic computation (verified with SymPy) yields

$$q'' + \widetilde{W}(y) q - k_x^2 q = \tilde{g}, \quad (7.2)$$

with

$$\widetilde{W}(y) = \frac{\rho_0''}{2\rho_0} - \frac{(\rho_0')^2}{4\rho_0^2}. \quad (7.3)$$

Difference from the original potential W : the $(\rho_0')^2/\rho_0^2$ coefficient is $-1/4$ rather than $-3/4$.

7.3 Surface-singularity comparison

For Lane–Emden $n = 3/2$, $\rho_0 \propto (R - y)^{3/2}$:

Formulation	Operator	Surface potential	$ C $
Original	$\nabla \cdot (\rho_0^{-1} \nabla p)$	$W \approx -21/(16t^2)$	1.3125
Reduced pressure	$\nabla \cdot (\rho_0 \nabla \pi)$	$\widetilde{W} \approx +3/(16t^2)$	0.1875

Singularity strength reduced by a factor of 7, and the sign of C flips from negative (attractive) to positive (repulsive). Frobenius indicial analysis:

Formulation	C	Indicial exponents	Square-integrable?
Original	$-21/16$	$7/4, -3/4$	only $\alpha = 7/4$ branch
Reduced pressure	$+3/16$	$3/4, 1/4$	both branches

This is a qualitative improvement: both linearly independent solutions are square-integrable, and SL theory applies directly.

Scope notice (2026-05-03 update). The factor-of-7 reduction documented above is meaningful only for $\sigma = 3/2$. For the project’s principal scenario $\sigma = 3$ (Eddington, where ρ_0 is a polynomial), raw Chebyshev collocation reaches machine precision directly, and this optimisation offers no operational value. See Part IV.

Part IV Direct Chebyshev Collocation (Phase 1 main route)

8. The integer-vs-fractional σ convergence cliff

The central finding of Phase 0 ext+ (2026-05-03): direct Chebyshev collocation of the reduced-pressure operator, with no substitution whatsoever, exhibits a **sharp dichotomy** in convergence behaviour depending on whether the surface exponent σ is integer or fractional.

8.1 Numerical evidence

Solve the manufactured-solution Poisson problem $\rho_0(r) = (1 - r)^\sigma$, $\pi_{\text{exact}} = \sin(2\pi r)$, $f = [\rho_0 \pi']' - k_x^2 \rho_0 \pi$ (with f constructed symbolically in SymPy).

N	$\sigma = 3$ error	$\sigma = 3/2$ error (raw)
16	8.5×10^{-8}	2.7×10^{-3}
32	$\sim 10^{-10}$	5.1×10^{-4}
64	6.7×10^{-11}	1.8×10^{-4}
128	$\sim 10^{-9}$ (roundoff)	5.2×10^{-5}
256	3.2×10^{-9}	1.1×10^{-5}

- $\sigma = 3$: exponential convergence to machine precision ($\sim 10^{-10}$ at $N = 64$), followed by a slow roundoff rise — the ideal behaviour of Chebyshev with a polynomial coefficient.
- $\sigma = 3/2$: steady N^{-2} algebraic convergence. No r^α prefactor recovers spectral accuracy.

8.2 Approximation-theoretic explanation (Trefethen Thm 7.2)

The decay of Chebyshev coefficients is governed by the analytic regularity of the function being expanded. For $\rho_0(r) = (1 - r)^\sigma$:

- $\sigma \in \mathbb{Z}_{\geq 0}$: ρ_0 is a **polynomial**, its Chebyshev expansion terminates in $\sigma + 1$ terms, and the whole coefficient is captured to machine precision. The spectral convergence of the outer problem is untouched by the surface.
- $\sigma \notin \mathbb{Z}$: $(1 - r)^\sigma$ has a **fractional algebraic branch point**. Its Chebyshev coefficients decay only as $N^{-\sigma-1/2}$. L_N approximates L only to this rate, and the rate propagates to the solution — hence N^{-2} for $\sigma = 3/2$.

8.3 Physical significance for stellar structure

The Lane–Emden equation $\theta'' + (2/\xi)\theta' + \theta^n = 0$ with $\theta(\xi_1) = 0$ has surface behaviour $\theta \sim (\xi_1 - \xi)$, so $\rho_0 \propto (\xi_1 - \xi)^n \propto (R - r)^n$. **The polytropic index n is literally the surface exponent σ .**

Index	Physical context	Surface σ	Rate
$n = 1$	outer white-dwarf layer	1	$N^{-3/2}$ algebraic
$n = 3/2$	convective core	$3/2$	N^{-2} algebraic
$n = 2$	main-sequence envelope	2	$N^{-5/2}$ algebraic
$n = 3$	Eddington radiative model	3	spectral
$n = 7/2$	giant hydrogen envelope	$7/2$	N^{-4} algebraic

The Eddington $n = 3$ model is the unique standard physical polytrope in the spectrally-convergent branch. This is a coincidence between “the most-studied historical polytrope” and “the best-behaved Chebyshev regime”, and it justifies choosing $n = 3$ as the background for the Phase 1 main route.

9. Chebyshev discretisation

9.1 Chebyshev–Gauss–Lobatto grid

$$\xi_j = \cos \frac{j\pi}{N}, \quad j = 0, \dots, N, \quad (9.1)$$

mapped affinely to $r \in [a, b]$. Let D denote the Trefethen spectral differentiation matrix of size $(N + 1) \times (N + 1)$.

9.2 Reduced-pressure operator

With $R_\rho \equiv \text{diag}(\rho_0(r_j))$,

$$L_N = D R_\rho D - k_x^2 R_\rho. \quad (9.2)$$

Dirichlet boundary conditions are imposed by strong collocation (unit rows at $j = 0$ and $j = N$, right-hand side set to prescribed values).

9.3 A symmetry caveat

D is **not symmetric** in the standard Euclidean inner product; it is symmetric in the Chebyshev–Gauss–Lobatto quadrature-weighted inner product. Eigenvalue problems must therefore be solved with `numpy.linalg.eig` (LAPACK `geev`), not `eigvalsh`. This was operationally important in Phase 0 ext+ Tests B and C: an initial implementation with `eigvalsh` produced spurious eigenvalues. See `docs/spectral_experiments.md` §19.

Part V Unified Basis Claim — Critical Examination

10. “g-mode as a free by-product” revisited

10.1 The two operators have different singular points

- **Poisson operator**: singular at the **surface** $r = R$ (where $\rho_0 \rightarrow 0$). Liouville potential $\widetilde{W} \sim \sigma(\sigma - 2)/[4(R - r)^2]$. The SymPy-derived optimal prefactor is

$$\pi = (R - r)^{\alpha_*} u, \quad \alpha_*(\text{Poisson}) = 1 - \sigma/2. \quad (10.1)$$

- **g-mode operator**: singular at the **origin** $r = 0$ (the centrifugal term $\ell(\ell + 1)/r^2$). The corresponding prefactor is

$$y_1 \sim r^{\beta_*}, \quad \beta_*(\text{g-mode}) = \ell + 1. \quad (10.2)$$

$\alpha_* \neq \beta_*$: **no single set of eigenfunctions simultaneously diagonalises both operators.**

10.2 The correct refined statement

The strong form of the original Liouville programme — “a unified SL basis simultaneously diagonalises the Poisson and g-mode operators” — fails. The weaker, operationally relevant form survives:

- The Chebyshev **mesh** is shared (same CGL nodes).
- The dense-linear-algebra infrastructure (cuBLAS GEMM, VRAM layout) is shared.
- The Poisson solve at each k_x reuses a single LU factorisation (only the scalar shift k_x^2 changes).
- The g-mode calculation is a **separate EVP on the same mesh**, not a free by-product.

The engineering benefit is preserved; the “free mathematical by-product” claim is demoted. This is the central conclusion of Phase 0 ext+ experiments E5 and E6.

Part VI Liouville Singularity — A Physical Interpretation

11. Reversed causality: quantum mechanics vs. astrophysics

The $1/t^2$ behaviour of the Liouville potential as $\rho_0 \rightarrow 0$ is mathematically identical to structures in quantum mechanics (the Coulomb potential, the centrifugal barrier). The **causal relationship**, however, is reversed.

11.1 In quantum mechanics: the potential is primary

$$V(r) \text{ (physical input)} \longrightarrow \psi(r) \rightarrow 0 \text{ (mathematical consequence)}$$

A Coulomb potential or centrifugal barrier is the **fundamental physical quantity**. The wavefunction ψ (and hence the probability density $|\psi|^2$) vanishes at certain boundaries *because* the potential forces it to. Techniques such as Coulomb–Sturmian bases ($r^\ell e^{-\alpha r} L_n$), Laguerre DVR, and R-matrix matching are designed around a singularity that **cannot be removed**, because it is a property of the physical system.

11.2 In astrophysics: the density is primary

$$\rho_0(y) \rightarrow 0 \text{ (physical input)} \longrightarrow W(y) \rightarrow -\infty \text{ (mathematical artifact)}$$

The density profile $\rho_0(y)$ is determined by hydrostatic equilibrium (Lane–Emden, MESA). The stellar surface is simply where the gas runs out; there is no “infinite potential barrier” in the physics. The original operator $\nabla \cdot (\rho_0^{-1} \nabla p)$ is a **degenerate elliptic** operator at the surface, losing ellipticity smoothly as $\rho_0 \rightarrow 0$, without any true singularity. The divergence of W is **manufactured by the $\sqrt{\rho_0}$ substitution** — it is a coordinate singularity, not a physical one.

11.3 Consequences for basis design

1. **Quantum-mechanics methods are applicable but over-engineered.** Coulomb–Sturmian bases were designed for genuine physical singularities; applying them to an artificial coordinate singularity is using a sledgehammer for a thumbtack.
2. **Path A’s limitations trace to the same observation.** A prefactor substitution r^α to absorb W ’s singularity is the astrophysical analogue of the Coulomb–Sturmian $r^\ell e^{-\alpha r}$ factor. There, it is the correct method; here, it attempts to “unwind” a singularity that ought not to exist in the first place.
3. **The natural approach is to return to the original operator.** Reduced pressure (Part III) weakens the singularity from non-integrable attractive to two-branch-integrable repulsive. Raw Chebyshev (Part IV) discretises the coefficient ρ_0 directly, and for polynomial σ encounters no singularity at all.

Summary. Compared with the technical machinery the quantum-mechanics community has developed for genuine Coulomb singularities, the stratified-Poisson problem in astrophysics should be handled with **minimal machinery**. For $\sigma \in \mathbb{Z}$, no special treatment is required; for $\sigma \notin \mathbb{Z}$, a Jacobi-weighted basis (Dedalus) is the only additional machinery needed.

Part VII Phase Roadmap

12. Current status and forward plan

12.1 Completed (Phase 0 ext+, 2026-05-02..03)

- Verified the convergence dichotomy between Lane–Emden $n = 3/2$ and $n = 3$.
- Benchmarked Chebyshev at $N = 48$ (192 DOF) against the GYRE full-gravity 4-variable system: max relative error 1.5×10^{-6} , a $21\times$ reduction in DOF and $350\times$ smaller max error compared with staggered FD at $N_r = 1024$ (4096 DOF).
- Three analytical-ceiling tests (manufactured Poisson, quantum harmonic oscillator, Dirichlet Laplacian) all reach double-precision machine accuracy (10^{-13} to 10^{-15}).
- Verified via barycentric Lagrange that “N coefficients $\neq$ N pixels”.
- Chose Path A (raw Chebyshev) as sufficient for $n = 3$.

Full evidence: docs/spectral_experiments.md and docs/spectral_stratified_poisson_report_2026-05-03.m

12.2 Phase 1: 2D Fourier–Chebyshev Boussinesq

- **x direction:** Fourier (periodic), reusing the `pseudo_spectral` cuFFT infrastructure.
- **y direction:** Chebyshev collocation, GPU-side $D^{(2)}$ by cuBLAS dense GEMM.
- **Physics:** 2D Boussinesq with buoyancy; Poisson via Chebyshev + dense solve.
- **Benchmark:** Rayleigh–Bénard Nu–Ra scaling (Ahlers et al.).
- **Background:** Gaussian transition, moving to Eddington $n = 3$ polytrope.

A design document `docs/phase1_2d_spectral_design.md` will be written when Phase 1 starts in earnest.

12.3 Phase 2: Anelastic upgrade

- Upgrade from Boussinesq to anelastic: $\nabla \cdot (\rho_0 \mathbf{\Gamma}) = 0$.
- Chebyshev handles the variable-density Poisson directly (raw or reduced-pressure form).
- **SL-GEMM backend as an optional optimisation:** activated only if dense Poisson solves become a GPU-wall-clock bottleneck.

12.4 Phase 3: Live eigenmode projection (differentiator)

- Independent EVP on the same mesh yields g-mode / p-mode eigenpairs.
- Instantaneous flow fields are projected onto the mode basis as a runtime diagnostic.
- **This is the genuine novelty of the project:** convection–pulsation coupling in a 2D nonlinear DNS with live modal projection.

12.5 Phase 4+: Spherical-shell extension

See `docs/sph_spectral_roadmap.md` (long-term).

12.6 Publication strategy

- **Methods paper** (JCP): “Sturm–Liouville spectral methods for stratified astrophysical flows — convergence regimes and GPU implementation”.
- **Applications paper** (A&C / ApJS): “GPU anelastic pseudo-spectral DNS with live eigenmode projection for convection–pulsation coupling diagnostics”.

The applications paper carries the real novelty; the methods paper functions as supporting citation.

Appendix A GPU implementation considerations

A.1 Batched GEMM formulation

Steps 3 and 5 of §5.2 are a single matrix–matrix multiplication:

$$\mathbf{\Gamma} = \Psi^\top \mathbf{\Gamma}, \quad \mathbf{\Gamma} = \Psi \mathbf{\Gamma},$$

with $\mathbf{\Gamma}, \mathbf{\Gamma}, \mathbf{\Gamma}, \mathbf{\Gamma} \in \mathbb{R}^{N_y \times N_x}$. On modern NVIDIA Ampere/Hopper GPUs, FP64 GEMM exceeds 1 TFLOP/s, and the $\mathcal{O}(N)$ arithmetic intensity (flops per byte read) ensures compute-bound execution for $N_y \geq 256$.

A.2 Integration with the time integrator

The SL-GEMM Poisson solver slots into an existing pseudo-spectral time integrator (e.g. IFRK3 + cuFFT) by replacing the spectral-space division $\hat{p} = -\hat{f}/|\mathbf{\Gamma}|^2$ with the sequence:

cuFFT (R2C in x) $\rightarrow$ cuBLAS DGEMM $\rightarrow$ pointwise divide $\rightarrow$ cuBLAS DGEMM $\rightarrow$ cuFFT (C2R in x).

All existing infrastructure — IFRK3 time integration, skew-symmetric convection, 2/3 dealiasing, VRAM frame buffering — is reused without modification.

A.3 Memory footprint

The transform matrix Ψ requires N_y^2 doubles = 32 MiB at $N_y = 2048$, which is modest compared with the flow-field arrays (~ 100 MiB each at 2048^2) and the VRAM frame buffer (~ 10 GiB).

Appendix B Related methods and literature positioning

B.1 The three-community gap

Community	Representative codes	Radial discretisation	Why SL-GEMM is absent
Stellar pseudo-spectral	ASH, Rayleigh	spherical harmonics (angular) + Chebyshev/FD (radial) + banded	GEMM not competitive on CPU
GPU pseudo-spectral	hit3d, spectralDNS	Fourier in all directions	limited to uniform density
Variable-density GPU CFD	engineering LES codes	finite volume + multigrid	spectral methods rarely used

No existing code combines Liouville reduction, SL spectral basis, GPU GEMM, and anelastic stellar convection.

B.2 SL theory background

- **Liouville normal form:** Sturm 1836, Liouville 1837; see Zettl (2005) for a modern treatment.
- **SL eigenfunction expansions for elliptic PDEs:** discussed by Boyd (2001) as theoretically elegant but rarely implemented due to the absence of a fast SL transform.
- **Chebyshev convergence theory:** Trefethen (2013) Ch. 7 (asymptotic coefficient decay for integer vs fractional branch points).
- **Berrut–Trefethen barycentric Lagrange interpolation** (2004): enables a stable, machine-precision evaluation of the spectral representation on any finer grid.

B.3 Relation to GYRE / Dedalus

See docs/singular_basis_survey_2026-05-02.md for a detailed survey. The project’s novelty positioning, in its final form, is “GPU 2D anelastic DNS with live eigenmode projection” — it does not compete with GYRE (the mature 1D stellar-pulsation code) or Dedalus (the general-purpose spectral-PDE framework) in the 1D pulsation-benchmark arena.

Appendix C Historical record — mapping to the four predecessor documents

Section in this document	Predecessor	Original section
Part I §1-2	anelastic_SL_spectral_design.md	§1-2
Part II §3-4	liouville_SL_spectral_derivation.md	§3-4
Part II §5	liouville_SL_spectral_derivation.md	§5
	§5 + anelastic_SL_spectral_design.md	
	§5	
Part III §6-7	reduced_pressure_liouville.md	entire
Part IV §8-9	polytropic_index_spectral_convergence_2026-05-03.md + formal report §3	
Part V §10	new (Phase 0 ext+ E5/E6 conclusions)	
Part VI §11	liouville_singularity_causality.md	entire
Part VII §12	new (post Phase 0 ext+ roadmap)	
Appendix A	anelastic_SL_spectral_design.md	§8 + liouville_SL_spectral_derivation.md
	§8	
Appendix B	anelastic_SL_spectral_design.md	§7 + liouville_SL_spectral_derivation.md
	§9	

All four predecessor files are retained in the repository as a record of the reasoning trajectory. Each carries a header update block pointing the reader to this document and to the formal report; new work should proceed from this document and the formal report only.

0. About this document

This document consolidates four experimental notes from 2026-05-02..03 into a single integrated record, reorganised around **setup** / **result** / **interpretation** for each experiment. The original four notes remain in the repository as an archival reasoning trajectory:

- docs/anelastic_sl_phase0_2026-05-02.md (§1-2 here: SL Poisson feasibility)
- docs/reduced_pressure_experiments_2026-05-02.md (§3-5 here: reduced-pressure validation)
- docs/gmode_experiments_2026-05-02.md (§6-15 here: the full g-mode experiment chain, Exp A through Exp K)
- docs/polytropic_index_spectral_convergence_2026-05-03.md (§16-19 here: the σ -dichotomy study)

Reproducibility protocol. Every experiment freezes its expected numerical output as an EXPECTED constant in the corresponding script. A `python scripts/<name>.py --verify` invocation must exit zero to confirm the numbers in the tables below. Any intentional change to these numbers must be landed in the same commit that updates both the EXPECTED and the corresponding table.

Sibling documents:

- docs/spectral_solver_design.md — design, derivations, roadmap.
- docs/spectral_stratified_poisson_report_2026-05-03.md — formal English report.
- docs/singular_basis_survey_2026-05-02.md — GYRE / Dedalus survey.

Part I SL Poisson feasibility (Phase 0, 2026-05-02)

1. Lane-Emden background and the Liouville potential W

1.1 Setup

The Emden equation $\theta'' + (2/\xi)\theta' + \theta^n = 0$ is integrated to the first zero ξ_1 .

1.2 Numerical results for Lane-Emden $n = 3/2$

Quantity	Value
ξ_1 (polytrope radius)	3.653754
ρ/ρ_c range	$[1.0, 2 \times 10^{-5}]$
$W(y)$ full range	$[-1.35 \times 10^6, -3.3]$
Surface region $\rho < 0.01$	$ W _{\max} = 1.35 \times 10^6$ (divergent)
Truncated domain $r/R_* \in [0, 0.94]$ ($\rho > 0.01$)	$W \in [-398, -3.3]$, bounded

Script: `scripts/anelastic_sl_phase0.py`.

1.3 Conclusion

The surface singularity exists as anticipated at the design stage. For $n = 3/2$ the safe truncation is $r/R_* \leq 0.94$ (the convective interior, excluding the thin surface atmosphere). This is the standard ASH / Rayleigh prescription.

2026-05-03 update: the truncation is required only for $n = 3/2$. For $n = 3$ (Eddington), $\rho_0 \propto (R - r)^3$ is a polynomial and Chebyshev handles it to machine precision without truncation (see Part IV).

2. SL eigenproblem + Fourier-limit degeneracy test

2.1 Discretisation

Interior finite differences on nodes $1, \dots, N - 2$ with implicit Dirichlet endpoints, $N = 512$. `scipy.sparse.linalg.eigsh` extracts the lowest 256 eigenvalues.

2.2 Fourier limit verification ($W = 0$)

With $W = 0$ the SL eigenvalues should reduce to $(n\pi/L)^2$.

n	μ_{SL}	$(n\pi/L)^2$	rel. err.
1	11.146	11.146	3.1×10^{-6}
20	4452.8	4458.4	1.26×10^{-3}

The $n = 20$ relative error matches the FD stencil theoretical limit $n^4 \pi^4 \Delta y^2 / (12L^4) \approx 1.26 \times 10^{-3}$ exactly. Measured = theory $\Rightarrow$ discretisation verified.

2.3 End-to-end SL-Poisson manufactured solution

N_{modes}	err_{L^2}	improvement
5	3.00×10^{-2}	—
10	7.34×10^{-3}	$4.1\times$
20	1.37×10^{-3}	$5.4\times$
40	2.18×10^{-4}	$6.3\times$
80	3.27×10^{-5}	$6.7\times$
160	5.99×10^{-6}	$5.5\times$
256	$\times 3.74010^{-6}$	converged to FD floor

The convergence slope $\log(\text{err})$ vs. $\log(N)$ is ≈ -3.5 , i.e. **algebraic**, not exponential. This is consistent with the theoretical expectation that the truncated surface $\rho \rightarrow 0$ should degrade the SL expansion from exponential to algebraic convergence.

2.4 Sturm oscillation + Tassoul asymptote (Phase 0 extensions E1, E2)

- **E1 (Sturm).** Each ψ_n has exactly n internal zeros (21 of 21 tested). This verifies bit-level correctness of the eigenvectors and preservation of topology.
- **E2 (Tassoul).** ΔP_n asymptotes to a constant with $\text{std}/\text{mean} = 8 \times 10^{-4}$. The shape matches Tassoul (1980) qualitatively, but a factor of 2.9 in magnitude remains — traceable to the Cowling slab approximation (which drops the $\ell(\ell + 1)/r^2$ centrifugal term).

2.5 Convergence order vs. cutoff (E3) — the smooth-vs-singular contrast

Scanning the Lane-Emden density threshold $\rho_{\text{threshold}}$:

cutoff	r_{hi}	err(256)	slope
0.1	0.77	4.3×10^{-7}	−2.42
0.01	0.94	3.7×10^{-6}	−2.39
0.001	0.99	2.7×10^{-5}	−2.26
0.0001	0.997	1.5×10^{-4}	−1.78

The closer to $\rho = 0$, the slower the convergence — the singularity’s influence is quantitatively visible.

Gaussian-capped smooth $\rho(y) = \exp(-2y^2) + 0.05$ (**no singularity**): $\text{err}(256) = 8.3 \times 10^{-7}$, semilog slope −0.049, giving $\text{err} \sim \exp(-0.05 N)$ — **exponential convergence**.

Publication-level summary:

“The SL method is exponentially convergent for smooth stratification. Algebraic convergence observed on Lane-Emden polytropes is entirely attributable to the surface singularity $\rho(R_\star) = 0$, quantifiable via cutoff scaling analysis.”

2.6 Brunt–Vaisala $N^2(r)$ vs. Liouville $W(r)$ — physical division of labour (E4)

Quantity	Encoding	Purpose
$W(r)$	pure density stratification (ρ'')	SL Poisson Liouville potential
$N^2(r)$	density + temperature (Schwarzschild)	g-mode frequencies / convective stability

Finding. Lane-Emden with γ -adiabatic law is Schwarzschild-neutral; $N^2 \equiv 0$ identically. A non-adiabatic perturbation ($\delta = 0.1 \sin(2\pi r)$ on top of T) gives $|N^2| \sim 1$, comparable in scale to $|W| \sim 10\text{--}400$.

2.7 Phase 0 cumulative verification table

Check	Status	Strength
Lane-Emden $W(y)$ singularity localisation	PASS	quantified via cutoff
SL discretisation Fourier limit	PASS	matches FD theoretical limit
Stable solution of first 256 eigenpairs	PASS	scipy eigsh, < 1 s
SL-Poisson $\text{err}_{L^2} = 3.7 \times 10^{-6}$	PASS	engineering-ready
Sturm oscillation (E1)	PASS	21/21 bit-correct
Tassoul asymptotic ΔP (E2)	PASS	asymptote attained
Exponential vs. algebraic (E3)	PASS	smooth $\Rightarrow$ exponential confirmed
$N^2 \leftrightarrow W$ division of labour (E4)	PASS	Phase 2/3 scope clarified

Phase 0 gate was originally PASS. Phase 0 ext+ subsequently demoted the “SL as optimal basis” angle to “same-mesh independent EVP” (see docs/spectral_solver_design.md Part V).

Part II Reduced-pressure follow-up (2026-05-02)

The experiments in this part numerically validate the theoretical prediction of docs/reduced_pressure_liouville.md — namely, that the reduced-pressure formulation weakens the surface singularity by a factor of 7. **Scope:** meaningful only for Lane-Emden $n = 3/2$ (fractional σ); not applicable to the project’s main $n = 3$ scenario (see Part IV).

3. Experiment A — breaking the Chebyshev floor

Script: scripts/reduced_pressure_chebyshev.py.

3.1 Motivation

The parent report’s §9 discretised the SL eigenvalue problem with finite differences, giving a floor $\sim 10^{-7}$ set by FD accuracy. A Chebyshev collocation eigensolver for (μ_n, ψ_n) breaks this floor and makes the two formulations quantitatively distinguishable.

3.2 Result (Lane-Emden $n = 3/2$, cutoff 0.01, N -mode scan)

N	err (original)	err (reduced-p)	ratio
5	6.0×10^{-3}	3.7×10^{-4}	16×
10	8.4×10^{-4}	6.9×10^{-5}	12×
20	7.8×10^{-5}	7.6×10^{-6}	10×
40	5.1×10^{-6}	5.7×10^{-7}	9×
80	3.1×10^{-7}	1.5×10^{-7}	2×
256	1.4×10^{-7}	1.5×10^{-7}	1× (FD floor)

Reduced-pressure is 10× more accurate in the spectral range $N \leq 40$; at high N both hit the FD floor.

3.3 Interpretation

- **Consistent $10\times$ advantage at low mode count.** GPU GEMM cost scales as N_y^2 ; using 20-40 SL modes is the natural engineering target, and this is exactly the range where reduced pressure excels.
- **Advantage vanishes for a smooth ρ .** With a Gaussian density profile (no singularity), the two formulations give identical convergence curves — confirming that the improvement stems from the weaker singularity, not from any generic property of the reduced-pressure variable.
- **The repulsive potential is better conditioned.** The original strongly attractive potential ($C = -21/16$) distorts low-order ψ_n toward the singular boundary; compensating requires many high-order modes. The weakly repulsive reduced-pressure potential ($C = +3/16$) leaves low-order ψ_n closer to Fourier modes, enabling rapid convergence at small mode count.

4. Experiment B — end-to-end Poisson convergence (on π , not q)

Script: `scripts/reduced_pressure_poisson_end2end.py`.

4.1 Setup

Manufactured $\pi_{\text{exact}}(x, y) = \sin(2\pi k_x x) \sin(\pi(y - y_{\text{lo}})/L)$ with $k_x = 2$. Forward and backward transport verified.

4.2 Principal result ($\rho_{\text{cut}} = 0.01$)

$\text{err}_{L^2}(\pi) = 1.8 \times 10^{-6}$ at $N = 256$ modes.

4.3 k_x -independence verification (Experiment C)

A single precomputed (μ_n, ψ_n) set attains comparable accuracy for every $k_x \in \{1, 2, 4, 8, 16\}$, confirming the theoretical prediction of §5.

5. Symbolic correction to the $\widetilde{W}$ coefficient

The parent report’s eq. (12) had a sign error (a missing factor of 3). The SymPy script `scripts/reduced_pressure_liouvi` computes, for $n = 3/2$,

$$\frac{1}{\sqrt{\rho_0}} \frac{d}{dy} \left[\rho_0 \frac{d}{dy} \left(\frac{q}{\sqrt{\rho_0}} \right) \right] = q'' + \frac{3}{16 t^2} q,$$

with $t = R - y$. This confirms $C = +3/16$, in agreement with §3. The parent report has been corrected.

Part III g-mode infrastructure and validation (Exps A-K)

The following is a condensed walk-through of the full g-mode experiment chain, from the initial incompressible-buoyancy simplification to the GYRE-compatible 4-variable adiabatic operator. Each entry is organised as **setup** / **result** / **conclusion**. Scripts are located in `scripts/gmode_exp_*.py`, with shared infrastructure in `scripts/gmode_infra.py`.

6. Exp A — Lane-Emden $\widetilde{W}$ -proxy heuristic

Script: `gmode_exp_a_lane_emden.py`, commit 8aa3476.

Setup. Lane-Emden $n = 3/2$, $\rho_{\text{cut}} = 0.05$, cavity $r \in [0.15, 0.844]$. Use $N_{\text{proxy}}^2 \equiv -\widetilde{W}(r)$ as input to the Cowling solver (30 radial orders, last-5 tail average).

Result. $\Delta P_{\text{tail}}/\Delta P_{\text{Tassoul}} = 0.852$ (within the acceptance window 0.80-1.20, PASS).

Conclusion. A pure pipeline smoke test. Lane-Emden itself is isentropic with $N^2 = 0$ and supports no true g-modes, but `solve_gmode_cowling` should reproduce the Tassoul asymptote whenever any positive N^2 profile is fed in.

7. Exp B — artificial Gaussian-bump N^2 convergence

Script: `gmode_exp_b_stratified.py`, commit 8aa3476.

Setup. Artificial Gaussian $N^2(r)$ on $[0.2, 1.0]$, $r_c = 0.6$, $\sigma = 0.2$, with a $\sin^2$ taper. Resolution scan $N_r \in \{256, 512, 1024, 2048\}$.

Result ($N_r = 2048$). $\Delta P_{\text{tail}}/\Delta P_{\text{Tassoul}} = 0.9993$; $|\text{ratio} - 1| = 7.5 \times 10^{-4}$. Convergence rate $\mathcal{O}(N_r^{-2})$, matching the second-order FD expectation.

Conclusion. The **first genuine g-mode calculation in the repository**. The infrastructure is ready for MESA profile ingestion, multi-cavity / radiative-convective boundary extensions, etc.

8. Exp C — Chebyshev collocation g-mode solver

Script: `gmode_exp_c_chebyshev.py`, commit e703991.

Setup. Same Gaussian bump as Exp B, now discretised with Chebyshev collocation (`solve_gmode_cowling_cheb`).

Result ($N_{\text{Cheb}} = 512$). $|\text{ratio} - 1| = 6.85 \times 10^{-5}$ — $4\times$ smaller DOF and $10\times$ smaller error than Exp B at $N_r = 2048$.

Spurious-mode guard. `n_modes = max(10, N_{\text{Cheb}}//5)` is enforced to avoid spectral-tail pollution. The Chebyshev D^2 is non-symmetric in the standard inner product, so one must use `numpy.linalg.eig` rather than `eigvalsh`. This lesson reappears in Phase 0 ext+ Tests B and C.

9. Exp D — polytropic profile via a MESA-style parser

Script: `gmode_exp_d_polytrope_profile.py`, commit c8b655c.

Setup. A MESA-style column-table reader ingests a synthetic $n = 3$ polytrope with a Gaussian N^2 fixture (600 rows), then hands it to the Cowling solver.

Result. Chebyshev $N = 512$ gives $|\text{ratio} - 1| = 8.1 \times 10^{-5}$. Drift relative to the in-memory Exp B/C numbers is 18%, dominated by the 600-row fixture interpolation error.

Conclusion. The parser pipeline is clean and can be swapped for a real `profile*.data`.

10. Exp E-G — 2-variable anelastic operator (transition phase)

Scripts: `gmode_exp_e_anelastic_linop.py`, `gmode_exp_f_variable_rho.py`, `gmode_exp_g_spherical_scalar.py`.

Setup. A 2-variable (y_1, p') anelastic operator; eliminating p' yields a scalar reduction. E/F/G cross-validate three algebraic forms.

Result. Exp E PASS on the Gaussian-bump cavity with the wide 0.85-1.20 acceptance window. Exp F PASS with variable density. Exp G spherical scalar vs. 2-variable PASS with $\mathcal{O}(N_r^{-2})$ convergence.

2026-05-02 corrections log. §7 originally described the scalar solver as the “Boussinesq limit of the 2-variable operator” — this was wrong. The scalar solver is the **slab / local-Cartesian** approximation (drops $\ell(\ell + 1)/r^2$), not a thermodynamic truncation (Boussinesq vs. anelastic). Consequently, the B-G PASS verdicts demonstrate **internal consistency**, not **external correctness**. The first external benchmark, Exp H, exposes this.

11. Exp H — first GYRE benchmark (exposing the problem)

Script: `gmode_exp_h_gyre_benchmark.py` plus `gmode_exp_h_run_gyre.sh`.

Setup. Build GYRE in the MESA SDK environment, run it on the bundled Lane-Emden $n = 3$ poly3 case, and compare against the Python 2-variable anelastic solver under full gravity.

Result. $n_g = 1$ ratio $2.2\times$ (120% disagreement); $n_g = 5$ ratio 1.3; $n_g = 10$ ratio 1.1; high- n_g ratio tends to 1 (the Boussinesq limit).

Diagnosis. Not a bug — a physics mismatch. `solve_anelastic_2var` uses only ρ_0, N^2 and **silently drops the V, U, Γ_1 coupling**. GYRE’s $\alpha_{\text{grv}} = 0$ pure Cowling 2-variable system requires five structure coefficients V, U, A^*, c_1, Γ_1 . Our solver is a Boussinesq-like simplification, not “anelastic in the stellar-oscillation sense”.

Conclusion. Re-implement the operator to match the GYRE equations (Exp I / Exp J).

12. Exp I — 2-variable Cowling GYRE-compatible (first external benchmark)

Script: `gmode_exp_i_gyre_compat.py`, commit 953d49f.

Setup. Implement GYRE’s $\alpha_{\text{grv}} = 0$ 2-variable Cowling equations, with $(y_1, y_2) = (x^{2-\ell}\xi_r/r, x^{2-\ell}P'/(pgr))$ and the five structure coefficients V, U, A^*, c_1, Γ_1 read from GYRE’s `poly3.txt`. Staggered FD at $N_r = 1024$, cutoffs `inner_cut = 0.01`, `outer_cut = 0.999`.

Result (vs. GYRE Cowling):

n_g	$\omega_{\text{GYRE}(\text{Cow})}^2$	ω_{ours}^2	rel. diff.
1	2.85195	2.85195	2.3×10^{-6}
2	1.36145	1.36146	3.1×10^{-6}
5	0.37473	0.37472	2.1×10^{-5}
10	0.11850	0.11843	5.6×10^{-4}

Max relative difference 5.6×10^{-4} , far below the 1% target. $n_g = 1$ **agrees to 5-6 significant figures**. The first apples-to-apples external benchmark PASS.

Cowling-limit quantified. GYRE Cowling vs. GYRE full differ by $\sim 13\%$ at $n_g = 1$ — the known Cowling approximation error (Unno et al. 1989).

13. Exp J — 4-variable full-gravity GYRE-compatible (FD production reference)

Script: `gmode_exp_j_full_gyre_compat.py`, commit `be94af9`.

Setup. Lift Cowling to $\alpha_{\text{grv}} = 1$ full 4-variable system (see `docs/spectral_solver_design.md` §4.2 and the formal report §4.2), adding $y_3 = \Phi'/(gr)$ and $y_4 = (d\Phi'/dr)/g$. Same FD grid.

Two bookkeeping bugs caught during implementation. An initial version gave 2.5% rel. diff. at $n_g = 1$ — physically untenable. Re-reading `A_t.inc` (GYRE stores the Jacobian transpose, so `A_t(i, j) = A(j, i)`) identified two errors:

1. Eq. 1 was missing the $\lambda/(c_1\omega^2)y_3$ term (present only when $\alpha_{\text{grv}} = 1$). This inverse- ω^2 term is what couples Φ' back into the displacement equation in the full-gravity case.
2. Eq. 2 had the y_3 coefficient as $-A^*$ instead of 0, and was missing the $-y_4$ term.

After correction the residual drops by four orders of magnitude:

n_g	$\omega_{\text{GYRE}(\text{full})}^2$	$\omega_{\text{ours}(\text{full})}^2$	rel. diff.
1	2.51593	2.51593	$\times 5.9010^{-7}$
2	1.28571	1.28571	2.7×10^{-5}
5	0.36993	0.36992	2.0×10^{-4}
10	0.11807	0.11801	5.3×10^{-4}

$n_g = 1$ **agrees to 6 significant figures**. This is the frozen FD production reference; its `--verify` regression is the oracle for any future CUDA port.

14. Exp K — Chebyshev 4-variable (spectral production)

Script: `gmode_exp_k_chebyshev_full.py`, commit `0da140f`.

Setup. Same equations and boundary conditions as Exp J, now discretised by Chebyshev collocation on $x \in [0.01, 0.999]$. `load_gyre_structure_interp_cheb` uses `scipy.interpolate.CubicSpline` (an initial implementation using `numpy.interp` hit a 3×10^{-5} floor; swapping to a cubic spline lowered it by four orders of magnitude to 8.7×10^{-9}).

Result ($N = 48, 192$ DOF):

n_g	ω_{GYRE}^2	$\omega_{\text{Chebyshev}}^2$	rel. diff.
1	2.51593	2.51593	5.9×10^{-7}
2	1.28571	1.28571	2.7×10^{-5}
5	0.36993	0.36992	2.0×10^{-4}
10	0.11807	0.11801	5.3×10^{-4}

Same accuracy with $21 \times$ **fewer DOF** and $350 \times$ **smaller maximum error** compared with Exp J at $N_r = 1024$.

15. Production-readiness summary

Operator	Method	$n_g = 1$ err vs. GYRE full
<code>'solve_gmode_cowling'(slab)</code>	FD, Boussinesq-like	$\sim 220\%$ (educational only)
<code>'solve_gmode_cowling_spherical'</code>	FD, scalar reduction	120%
<code>'solve_anelastic2var'</code>	FD, no $V/U/\Gamma_1$	120%
<code>'solve_gmode_cowling_gyre_compat'(2-var)</code>	FD, Cowling	13.4% (Cowling limit)
<code>'solve_gmode_full_gyre_compat'(4-var)</code>	FD, full gravity	$\times 5.9010^{-7}$ (FD production ref)
<code>'solve_gmode_full_chebyshev'(Exp K)</code>	Chebyshev, full	$\times 5.9010^{-7}$ (spectral production, $21 \times$ less DOF)

Part IV Polytropic-index convergence dichotomy

Background. Phase 0 (Part I) observed $N^{-2.4}$ algebraic convergence on Lane-Emden. A systematic Phase 0 ext+ scan reveals that the convergence order depends sharply on whether the surface exponent σ is integer or fractional, with clear consequences for the project's choice of background polytrope.

16. E6 v2 convergence scan (SymPy-forced manufactured solution)

Script: `scripts/spectral_liouville_convergence_v2.py`.

16.1 Setup

$\rho(r) = (1 - r)^\sigma$ on $[0, 1]$; $\pi_{\text{exact}} = \sin(2\pi r)$ (Dirichlet-compatible); the forcing $f = [\rho\pi']' - k_x^2\rho\pi$ is constructed symbolically in SymPy and evaluated on the CGL grid. Dirichlet BCs $\pi(0) = \pi(R) = 0$.

16.2 $\sigma = 3$ (Lane-Emden $n = 3$)

N	err (raw)	err ($\alpha = 1 - \sigma/2 = -1/2$)
16	8.5×10^{-8}	2.4×10^{-3}
32	$\sim 10^{-10}$	$\sim 10^{-4}$
64	6.7×10^{-11}	1.5×10^{-4}
128	$\sim 10^{-9}$ (roundoff)	$\sim 10^{-5}$
256	3.2×10^{-9}	9.1×10^{-6}

Raw Chebyshev reaches machine precision. Adding a prefactor makes the error **worse** (fractional α introduces algebraic irregularity at the endpoint).

16.3 $\sigma = 3/2$ (Lane-Emden $n = 3/2$)

N	err (raw)	$\alpha = 1/4$	$\alpha = -1/2$	$\alpha = -3/4$
16	2.7×10^{-3}	1.1×10^{-2}	1.3×10^{-2}	7.8×10^{-2}
64	1.8×10^{-4}	7.5×10^{-4}	1.1×10^{-3}	2.1×10^{-2}
256	1.1×10^{-5}	4.8×10^{-5}	8.3×10^{-5}	5.4×10^{-3}

$N^{-2.0}$ **algebraic convergence for all α .** No simple power-law prefactor restores spectral accuracy.

17. Why integer vs. fractional σ matters

17.1 Approximation-theoretic view

The decay of Chebyshev coefficients is governed by analytic regularity (Trefethen 2013, Thm. 7.2):

- $\rho(r) = (R - r)^3 = R^3 - 3R^2r + 3Rr^2 - r^3$ is a **polynomial**, with a finite Chebyshev expansion (4 terms). The smoothness of $\rho(r)\pi(r)$ is inherited entirely from $\pi(r)$.
- $\rho(r) = (R - r)^{3/2}$ is analytic on $[0, R)$ but has a branch point at R . Its Chebyshev coefficients decay only as $N^{-\sigma-1/2} \sim N^{-2}$.

The N^{-2} convergence is a direct consequence of Chebyshev's inability to resolve a fractional-power branch point with exponential accuracy.

17.2 Physical significance (Lane-Emden surface structure)

The Lane-Emden equation $\theta'' + (2/\xi)\theta' + \theta^n = 0$ with $\theta(\xi_1) = 0$ has surface behaviour $\theta(\xi) \sim (\xi_1 - \xi)$, so $\rho \propto (\xi_1 - \xi)^n \propto (R - r)^n$.

The polytropic index n is literally the surface exponent σ . See the table in docs/spectral_solver_design.md Part IV §8.3.

18. What actually works for fractional σ

1. **Jacobi-weighted basis** (Dedalus). Expand π in $\{(1 - r)^\sigma J_n^{(\sigma,0)}(r)\}$ — the basis carries the singular behaviour, and the coefficient expansion is over polynomials. **Gives spectral convergence for any $\sigma > -1$.**
2. **Coordinate stretching** (Kosloff–Tal-Ezer). Introduce a change of variable $r = r(s)$ that concentrates integration near the surface. Used neither by GYRE nor by Dedalus.

Both options lie **outside the Liouville framework**.

19. Analytical ceiling tests (E7b)

Script: `scripts/spectral_analytical_ceiling.py`.

Motivation. Does the 8.7×10^{-9} floor of Exp K represent the Chebyshev ceiling itself, or is it a property of the GYRE input data? Analytical-solution tests discriminate.

19.1 Test A — manufactured Poisson (reused from §16)

$\sigma = 3$ with SymPy-exact forcing. Reaches 5.5×10^{-13} at $N = 24$ — essentially double-precision rounding.

19.2 Test B — quantum harmonic oscillator on $[-10, 10]$

$-\psi'' + x^2\psi = \lambda\psi$, exact eigenvalues $\lambda_n = 2n + 1$. The first five eigenvalues reach 3×10^{-13} relative error at $N = 64$.

19.3 Test C — Dirichlet Laplacian on $[0, 1]$

$-u'' = \lambda u$, exact eigenvalues $\lambda_n = n^2\pi^2$. $N = 16$ already reaches 10^{-14} .

19.4 A key lesson — non-symmetry of D^2

An initial implementation of Tests B and C with `np.linalg.eigvalsh` gave divergent / spurious eigenvalues. The Chebyshev D^2 is not symmetric in the Euclidean inner product; one must use `np.linalg.eig` and filter for finite, positive, real-part eigenvalues. The lesson is in the literature but easy to forget.

19.5 Conclusion

Exp K's 8.7×10^{-9} floor is not a spectral limit. It is the precision ceiling of GYRE's 999-point `poly3.txt` input. Recovering more precision would require rebuilding the polytrope with a GL6 integrator at $\sim 10^4$ nodes and relative tolerance 10^{-14} .

Part V Barycentric interpolation — resolution vs. representation

20. The spectral representation is a continuous function

Script: `scripts/spectral_resolution_demo.py`.

20.1 Concept

The $N + 1$ Chebyshev coefficients define a **continuous function** evaluable at any point by barycentric Lagrange interpolation (Berrut & Trefethen 2004, SIAM Rev 46, 501-517):

$$u(r^\star) = \frac{\sum_j w_j u_j / (r^\star - r_j)}{\sum_j w_j / (r^\star - r_j)}, \quad w_j = (-1)^j c_j,$$

with $c_0 = c_N = 1/2$, $c_j = 1$ otherwise. $\mathcal{O}(N)$ per evaluation point, stable against catastrophic cancellation, error $\leq 10^{-12}$.

20.2 Numerical verification

The Exp K $N = 48$ (49 CGL nodes) representation, barycentric-evaluated at 4096 points, vs. the Exp J FD $N_r = 1024$ solution cubic-spline interpolated at the same 4096 points: max diff 3.4×10^{-3} , **exactly the Exp K discretisation error**; the barycentric contribution is $< 10^{-12}$.

20.3 Implication for 2D simulations

A pseudo-spectral 2048^2 field can be **rendered at 4K / 8K resolution without loss**. The real resolution ceiling is the 2/3 dealiasing cutoff ($2N/3 \approx 1365^2$), not the grid itself.

Part VI Frozen regression assets

21. Frozen 1D g-mode operators

Operator	Method	Accuracy vs. GYRE	Role
<code>solve_gmode_full_gyre_compat</code> (<code>gmode_infra.py</code>)	Staggered FD $N_r = 1024$	$n_g = 1$ 5.9×10^{-7} , max 5.3×10^{-4}	FD regression oracle
<code>solve_gmode_cowling_gyre_compat</code> (<code>gmode_infra.py</code>)	Staggered FD $N_r = 1024$	vs. GYRE Cowling 5.6×10^{-4}	FD Cowling baseline
<code>solve_gmode_full_chebyshev</code> (<code>gmode_exp_k_chebyshev_full.py</code>)	Chebyshev $N = 48$	max 1.5×10^{-6}	Chebyshev production ref.

22. Analytical-verification scripts (frozen EXPECTED)

- `gmode_exp_i_gyre_compat.py --verify` — 2-var Cowling, max 5.6×10^{-4} .
- `gmode_exp_j_full_gyre_compat.py --verify` — 4-var full, max 5.3×10^{-4} .
- `gmode_exp_k_chebyshev_full.py --verify` — Chebyshev 4-var, max 1.5×10^{-6} .
- `spectral_analytical_ceiling.py` — three ceiling tests, all 10^{-13} to 10^{-15} .

23. Exploratory scripts (retained, not to be modified)

- `gmode_exp_a..g*.py` — educational baselines (Boussinesq-like simplifications).
- `gmode_exp_h*.py` — the first GYRE benchmark, diagnostic record.
- `reduced_pressure*.py` — Liouville-form validation suite.
- `anelastic_sl_phase0*.py` — Lane-Emden $n = 3/2$ Phase 0.
- `spectral_liouville_beta_derivation.py` — SymPy α_* .
- `spectral_liouville_convergence_v2.py` — σ -dichotomy data.
- `spectral_liouville_prefactor.py` — Path A α sweep.
- `spectral_resolution_demo.py` — barycentric demo.

Appendix A Five central findings of Phase 0 ext+ (F1-F5)

The following five points, surfaced during the Phase 0 ext+ conversation, extend beyond the tabulated experimental data:

- **F1.** Exp K’s 8.7×10^{-9} floor is not a spectral limit but the GYRE 999-point input precision ceiling. Verified by the analytical-ceiling tests.
- **F2.** “ N coefficients $\neq N$ pixels.” The $N + 1$ Chebyshev coefficients define a continuous function evaluable at arbitrary resolution via barycentric Lagrange interpolation.
- **F3.** Gibbs vs. stellar pulsation. Low-frequency linear problems reach machine precision with spectral methods; strong-shock / supersonic flows do not (switch to `cart_ale2`). For the stellar2d use case (pulsation + weak-to-moderate convection + smooth polytropic background), the practical spectral ceiling is far above 2D FD.
- **F4.** Revised project positioning. Novelty is not in 1D stellar pulsation (GYRE / Reese–Lignières / Dedalus occupy that territory) but in **2D GPU DNS with live eigenmode projection**.
- **F5.** The strong form of “unified basis simultaneously diagonalises Poisson and g-mode” fails ($\alpha_* \neq \beta_*$); what survives is “same-mesh independent EVP”.

Appendix B Mapping to the four predecessor documents

Section in this document	Predecessor	Original section
Part I §1-2	<code>anelastic_sl_phase0_2026-entire</code> 05-02.md	
Part II §3-5	<code>reduced_pressure_experiments_2026-entire</code> 05-02.md	
Part III §6-14	<code>gmode_experiments_2026-§2-15</code> 05-02.md	
Part III §15	<code>gmode_experiments_2026-§15 production readiness</code> 05-02.md	
Part IV §16-19	<code>polytropic_index_spectral_Fcn</code> §16 convergence 05-03.md + <code>phase0_ext_plus_summary_2026-</code> 05-03.md	

Section in this document	Predecessor	Original section
Part V §20	phase0_ext_plus_summary_F2026-05-03.md	(2026 resolution)
Part VI §21-23	phase0_ext_plus_summary_F2026-05-03.md	(2026 assets)
Appendix A	phase0_ext_plus_summary_F2026-05-03.md	(2026-)

All four predecessor files remain in the repository as a reasoning trajectory. Each carries an update block pointing here and to the formal report; new work should be framed against this document and the formal report.